

EX.NO : 2	ENCRYPTION CRYPTO 101 IN TRYHACKME PLATFORM
DATE:	

AIM : To understand the fundamentals of **cryptography** by performing hands-on exercises in the **Crypto 101** room on the **TryHackMe** platform, including basic encryption, encoding, hashing, and cryptographic concepts

TOOLS

- Web browser (Chrome / Firefox)
- Internet connection
- TryHackMe account (free)
- Built-in TryHackMe terminal (AttackBox) or local Linux machine

THEORY

Cryptography is the practice of securing information by converting plain text into unreadable format (cipher text) using encryption techniques.

Crypto 101 introduces basic concepts such as:

- Encoding vs Encryption
- Symmetric and Asymmetric encryption
- Hashing
- Common ciphers (Caesar, Base64, etc.)

STEPS TO PERFORM THE EXPERIMENT

Step 1: Login to TryHackMe

1. Open browser and go to: <https://tryhackme.com>
2. Login using your registered credentials.

Step 2: Open Crypto 101 Room

1. In the search bar, type Crypto 101
2. Click on the room named Crypto 101
3. Click Join Room

Step 3: Start the Machine (if required)

1. Click Start Machine / Start AttackBox
2. Wait for the virtual environment to load.

Step 4: Perform Cryptography Tasks

Complete the tasks provided in the room, such as:

4.1 Encoding & Decoding

(a) Base64 Encoding / Decoding

Purpose:

Base64 is an encoding technique used to represent binary data in readable ASCII format.

How to Solve

To Decode Base64

1. Copy the given Base64 string from the question.
2. Open the TryHackMe terminal.
3. Use the command:
`echo "Q3J5cHRvZ3JhcGh5" | base64 -d`
4. The output will be the decoded plain text.
5. Enter the decoded text into the answer box.

To Encode Base64

```
echo "cryptography" | base64
```

Expected Output Example: Cryptography

(b) Hexadecimal Conversion

Purpose:

Hexadecimal is a base-16 encoding system commonly used in computing.

How to Solve

Hex to Text

```
echo 48656c6c6f | xxd -r -p
```

Text to Hex

```
echo "Hello" | xxd -p
```

Expected Output: Hello

4.2 Classical Ciphers

(a) Caesar Cipher

Purpose:

Caesar cipher shifts letters by a fixed number.

How to Solve

1. Identify the shift value (given or by trial).
2. Use online Caesar decoder or terminal tool.

Using terminal

```
echo "KHOOR" | tr 'A-Z' 'X-ZA-W'
```

(Shift = 3)

Expected Output: HELLO

(b) Shift-Based Encryption

Purpose:

Similar to Caesar cipher but may use different shift values.

How to Solve

1. Try different shift values until meaningful text appears.
2. Online tools can be used if allowed.

Example:

- Shift 1 → incorrect
- Shift 3 → readable text → correct answer

RESULT : Thus the Encryption 101 room has been successfully completed.